

BOUNDARY DEHN TWISTS ARE OFTEN COMMUTATORS

AYODEJI LINDBLAD

ABSTRACT. For X any complete intersection of even complex dimension or any connected sum thereof (or, more generally, any space among certain broad classes of smooth manifolds), we concretely construct orientation-preserving diffeomorphisms a, c of punctured X rel boundary whose commutator $[a, c]$ represents the smooth mapping class rel boundary of the boundary Dehn twist. This shows that boundary Dehn twists on 4-manifolds known to be nontrivial in the smooth mapping class group rel boundary by work of Baraglia-Konno, Kronheimer-Mrowka, J. Lin, and Tilton become trivial after abelianization, generalizing work of Y. Lin which applied an argument based on the global Torelli theorem and an obstruction of Baraglia-Konno to prove that the abelianized boundary Dehn twist on the punctured $K3$ surface is trivial.

1. BACKGROUND AND MAIN RESULTS

For a smooth oriented manifold X , we consider the groups $\text{Diff}^+(X)$ of orientation-preserving diffeomorphisms of X and $\pi_0(\text{Diff}^+(X))$ of smooth isotopy classes of $\text{Diff}^+(X)$, called the *smooth mapping class group of X* . When X is closed, we denote by X° the complement of an open ball in X and consider the groups $\text{Diff}^+(X^\circ, \partial)$ of elements of $\text{Diff}^+(X^\circ)$ which restrict to the identity in a neighborhood of the boundary and $\pi_0(\text{Diff}^+(X^\circ, \partial))$ of smooth isotopy classes of $\text{Diff}^+(X^\circ, \partial)$, called the *smooth mapping class group rel boundary of X°* . For $[\alpha]$ a generator of $\pi_1(SO(n))$, a diffeomorphism of X° taking $(t, \omega) \mapsto (t, \alpha(t)\omega)$ on a collar neighborhood $\nu(\partial X^\circ) \cong [0, 1] \times S^{n-1}$ of ∂X° and restricting to the identity elsewhere has smooth mapping class rel boundary $t_X \in \pi_0(\text{Diff}^+(X^\circ, \partial))$ which is referred to as the *boundary Dehn twist on X°* . Denoting by $\text{BDiff}^+(X^\circ, \partial)$ the classifying space of $\text{Diff}^+(X^\circ, \partial)$, the association $\pi_0(\text{Diff}^+(X^\circ, \partial)) \cong \pi_1(\text{BDiff}^+(X^\circ, \partial))$ gives rise to the abelianization map

$$\cdot^{ab} : \pi_0(\text{Diff}^+(X^\circ, \partial)) \rightarrow H_1(\text{BDiff}^+(X^\circ, \partial); \mathbb{Z}),$$

allowing us to consider the *abelianized boundary Dehn twist t_X^{ab} on X°* . The study of boundary Dehn twists and their abelianizations is then motivated by the recent wave of interest in boundary Dehn twists on 4-manifolds [3, 5, 14, 17, 18, 19, 22, 24, 25, 33, 35] alongside the sustained interest in

abelianizations of mapping class groups [6, 9, 20, 21, 23, 31, 34] and cohomology classes of $\mathrm{BDiff}^+(X^\circ, \partial)$ [8, 10, 12, 15, 16, 27, 28, 30, 32], which correspond to characteristic classes of smooth X -bundles. For a simply-connected closed 4-manifold X , Y. Lin asked whether the abelianized boundary Dehn twist on X° is trivial and observed [26, Proposition 2.1] that such triviality is equivalent to the existence of a smooth orientable X -bundle over an orientable closed surface whose total space is not spin. Y. Lin then applied an argument based on the global Torelli theorem (via an argument [4, Theorem 1.1] of Baraglia and Konno) and an obstruction [3, Proposition 4.21] of Baraglia and Konno to prove existence of such a bundle when $X = K3$, answering this case of the question in the affirmative. We observed the following concrete reproof of this result: viewing the $K3$ surface as the complex hypersurface of $\mathbb{P}^3$ cut out by the polynomial $z_0^4 + z_1^4 + z_2^4 + z_2 z_3^3$, the involutions of $\mathbb{P}^3$ induced by the antipodal map in the zeroth complex coordinate and complex conjugation restrict to commuting smooth involutions $a, c \in \mathrm{Diff}^+(K3)$ with a shared fixed point $[0 : 0 : 0 : 1] \in K3$, at which a and c admit certain local behavior of interest. From such involutions (whose mapping torus will have non-spin total space), we may then construct diffeomorphisms $a^\circ, c^\circ \in \mathrm{Diff}^+(K3^\circ, \partial)$ whose commutator $[a^\circ, c^\circ]$ represents the smooth mapping class rel boundary of the boundary Dehn twist on $K3^\circ$. Underlying this observation is a general concrete construction of such commuting smooth involutions on all smooth complete intersections (by which we mean complex $(n - m)$ -manifolds cut out in $\mathbb{P}^n$ by $m < n$ homogeneous polynomials) of even complex dimension, on all connected sums thereof, and on further broad classes of smooth manifolds; we formalize this construction to provide a broadly applicable concrete realization of the boundary Dehn twist as a commutator in the smooth mapping class group rel boundary. This is communicated by Theorem 1.1, which then has the consequence that the abelianized boundary Dehn twist is often trivial, even on 4-manifolds on which it is known—by work of Baraglia and Konno [3, 5], Kronheimer and Mrowka [24], J. Lin [25], and Tilton [35]—that the boundary Dehn twist is nontrivial in the smooth mapping class group rel boundary.

Theorem 1.1 (Main Theorem). *Let X be any smooth complete intersection of even complex dimension or any connected sum thereof (or any other space admitting $a, c \in \mathrm{Diff}^+(X)$ as in Proposition 2.1; further classes of such spaces are presented at the end of this section). We may construct $a^\circ, c^\circ \in \mathrm{Diff}^+(X^\circ, \partial)$ whose commutator $[a^\circ, c^\circ]$ represents the smooth mapping class rel boundary t_X of the boundary Dehn twist on X° (showing that the abelianized boundary Dehn twist t_X^{ab} is trivial). Moreover, we may construct a smooth orientable X -bundle over T^2 whose total space is not spin.*

There has been substantial interest in recent years in identifying, for a simply-connected closed 4-manifold X , whether the boundary Dehn twist on X° is nontrivial in the smooth mapping class group rel boundary (in

which case it is a *relatively exotic diffeomorphism* [33, Theorem E]). To the knowledge of the author, this nontriviality has only been shown for X homeomorphic to the $K3$ surface by work of Baraglia and Konno [3, Corollary 1.3] and Kronheimer and Mrowka [24, Proposition 1.2], for X the once-stabilized $K3$ surface $K3 \# (S^2 \times S^2)$ by work [25] of J. Lin, for any X satisfying a technical condition—principal examples of which include a large collection of complete intersections and double logarithmic transforms of the elliptic surfaces $E(4n - 2)$ [5, Theorem 1.5]—by other work [5, Theorem 1.4] of Baraglia and Konno, and for X the connected sum $K3 \# K3$ of two $K3$ surfaces by work [35] of Tilton. The $K3$ surface, the once-stabilized $K3$ surface, the connected sum of two $K3$ surfaces, and the infinitely many complete intersections satisfying the aforementioned technical condition are all smooth complete intersections or connected sums thereof, so Theorem 1.1 gives a concrete expression for the nontrivial boundary Dehn twists on their punctures as a commutator in the smooth mapping class group rel boundary , showing that they become trivial after abelianization (a fact shown in the setting of the $K3$ surface by Y. Lin [26]). Moreover, note that the elliptic surfaces $E(n)$ are natural candidates for future study of boundary Dehn twists for n even (as this is when $E(n)$ is spin; the boundary Dehn twist is trivial on the puncture of any non-spin simply-connected closed manifold [33, Corollary A.5]). As observed in Section 4, these spaces belong to another broad class of spaces X for which we verify Theorem 1.1.

The three sections to follow are each dedicated to expositing and proving a central proposition of interest. Proposition 2.1 states that we may build $a^\circ, c^\circ \in \text{Diff}^+(X^\circ, \partial)$ whose commutator $[a^\circ, c^\circ]$ realizes the smooth mapping class rel boundary of the boundary Dehn twist on X° from $a, c \in \text{Diff}^+(X)$ satisfying a certain straightforward condition (whose mapping torus we also show will not be spin). Note that we may construct a pair of orientation-preserving diffeomorphisms satisfying this condition on $X_1 \# X_2$ for any manifolds X_1, X_2 of the same dimension at least 4 each admitting such a pair. Proposition 3.1 then states that we may construct such a pair on any complete intersection of even complex dimension (in a complex projective space), while Proposition 4.1 gives the analogous statement for a more general class of complete intersections of even complex dimension in products of complex projective spaces. Our proofs of these propositions imply that these spaces are typically holomorphic double branched covers with compatible real structures whose branch divisors contain a real point; in fact, the deck transformation and real structure of any such double branched cover of even complex dimension also give such a pair. For any 2-fold connected sum of a double branched cover of (real) dimension at least 3, we may arrange that the same is true of the natural extension of the deck transformation to the connected sum and the map exchanging the connected summands. This verifies Theorem 1.1 when X is any of these spaces or any connected sum thereof. We omit further examples of spaces to which the theorem applies

for brevity and invite the reader to consider settings which may be of interest. In light of the generality of the theorem, we conclude by restating the motivating question [26] posed by Y. Lin with a more pointed framing:

Question 1.2. Does there exist any smooth simply-connected closed 4-manifold X for which the abelianized boundary Dehn twist t_X^{ab} is nontrivial?

2. BOUNDARY DEHN TWISTS AS COMMUTATORS

In this section, we present criteria for a pair of orientation-preserving diffeomorphisms of an oriented closed manifold to give rise to diffeomorphisms rel boundary of the puncture of this manifold whose commutator represents the smooth mapping class rel boundary of the boundary Dehn twist. Moreover, this criterion ensures that the mapping torus of these diffeomorphisms is not spin. These results are communicated by Proposition 2.1.

Proposition 2.1. *Consider any smooth oriented closed manifold X and $a, c \in \text{Diff}^+(X)$ whose commutator $[a, c]$ is smoothly isotopic to the identity through diffeomorphisms whose differentials each act as the identity at a shared fixed point $x \in X$ of a and c . Say the dimension of the maximal subspace of $T_x X$ on which all generalized eigenvalues of the differentials da_x and dc_x are negative real numbers is odd. We may construct $a^\circ, c^\circ \in \text{Diff}^+(X^\circ, \partial)$ whose commutator $[a^\circ, c^\circ]$ represents the smooth mapping class rel boundary of the boundary Dehn twist on X° . Moreover, the mapping torus $X_{a,c} \rightarrow T^2$ of a and c is a smooth orientable X -bundle over T^2 whose total space is not spin.*

We discuss preliminary facts central to the proofs throughout this section in Subsection 2.1. In Subsection 2.2, we produce $a', c' \in \text{Diff}^+(X)$ admitting certain standard local behavior in a neighborhood of x from a, c as in Proposition 2.1, then discuss why the connected sum $X_1 \# X_2$ of any manifolds X_1, X_2 of the same dimension at least 4 each admitting diffeomorphisms as in Proposition 2.1 will also admit such diffeomorphisms. In Subsection 2.3, we construct $a^\circ, c^\circ \in \text{Diff}^+(X^\circ, \partial)$ from a', c' and show that their commutator represents the smooth mapping class rel boundary of the boundary Dehn twist. Following this, we conclude the proof of the proposition by showing that the mapping torus $X_{a,c} \rightarrow T^2$ is not spin in Subsection 2.4. Finally, we discuss approaches to represent the mapping class of the boundary Dehn twist as a commutator in groups other than $\text{Diff}^+(X^\circ, \partial)$ in Subsection 2.5.

2.1. Paths of matrices. Writing

$$(1) \quad r_k := \begin{bmatrix} -1 & & \\ & -1 & \\ & & 1 \end{bmatrix}, \quad r_i := \begin{bmatrix} 1 & & \\ & -1 & \\ & & -1 \end{bmatrix},$$

we first define paths in $SO(n)$ (for $n \geq 3$) taking the identity to $r_k \oplus I_{n-3}$ and $r_i \oplus I_{n-3}$, whose commutator loop Remark 2.2 observes will represent the nontrivial element of $\pi_1(SO(n))$. For commuting $A, C \in GL^+(n)$, we then observe—as communicated by Remark 2.3—that we may produce smooth paths through commuting matrices respectively taking A and C to $r_k^\nu \oplus I_{n-3}$ and $r_i^\nu \oplus I_{n-3}$, where $\nu \in \{0, 1\}$ is the parity of the dimension of the maximal subspace of $\mathbb{R}^n$ on which all generalized eigenvalues of A and C are negative real numbers. To make the former observation, define

$$(2) \quad R_k(t) := \begin{bmatrix} \cos(\pi t) & -\sin(\pi t) & \\ \sin(\pi t) & \cos(\pi t) & \\ & & 1 \end{bmatrix}, \quad R_i(t) := \begin{bmatrix} 1 & & \\ & \cos(\pi t) & \sin(\pi t) \\ & -\sin(\pi t) & \cos(\pi t) \end{bmatrix}$$

for $t \in [0, 1]$, so we have

$$R_k(0) = R_i(0) = I_3, \quad R_k(1) = r_k, \quad R_i(1) = r_i.$$

Denoting by

$$\mathbb{H} := \mathbb{R}\langle i, j \rangle / (i^2 = j^2 = -1, ij = -ji)$$

the quaternions, by $\text{Spin}(3)$ the group of unit quaternions, and by

$$\text{Im}(\mathbb{H}) := \mathbb{R}i \oplus \mathbb{R}j \oplus \mathbb{R}k \quad (k := ij)$$

the vector space of purely imaginary quaternions, the mapping

$$\pi : \text{Spin}(3) \rightarrow \text{SO}(3) \cong \text{SO}(\text{Im}(\mathbb{H})), \quad q \mapsto (v \mapsto q^{-1}vq)$$

is the universal cover of $\text{SO}(3)$. The paths in $\text{Spin}(3)$ defined by

$$Q_k(t) := e^{\pi kt/2}, \quad Q_i(t) := e^{\pi it/2}$$

for $t \in [0, 1]$ are respectively lifts of the paths R_k, R_i in $\text{SO}(3)$, so the commutator path $[Q_k, Q_i]$ is a lift of $[R_k, R_i]$. Now, we have

$$[Q_k(1), Q_i(1)] = ki(-k)(-i) = -1 \neq 1 = [Q_k(0), Q_i(0)],$$

so the commutator loop $[R_k, R_i]$ must represent the nontrivial element of $\pi_1(SO(3)) \cong \mathbb{Z}/2\mathbb{Z}$. Fixing $n \geq 3$ and noting that the inclusion $\mathbb{R}^3 \hookrightarrow \mathbb{R}^n$ into the first three factors induces an isomorphism $\pi_1(SO(3)) \cong \pi_1(SO(n))$, we then get exactly the result of Remark 2.2:

Remark 2.2. For $n \geq 3$, the commutator loop $[R_k \oplus I_{n-3}, R_i \oplus I_{n-3}]$ represents the nontrivial element of $\pi_1(\text{SO}(n))$.

For $\mathcal{R} \subset GL^+(n) \times GL^+(n)$ the set of commuting pairs in $GL^+(n)$ and $(A, C) \in \mathcal{R}$, the Jordan-Chevalley decomposition gives rise to a smooth path $\rho_S : [0, 1] \rightarrow \mathcal{R}$ from (A, C) to a pair $\rho_S(1)$ of semisimple matrices with eigenvalues and associated eigenspaces the generalized eigenvalues and associated generalized eigenspaces of (A, C) . Rescaling these matrices on their common eigenspaces, we can produce a smooth path $\rho_N : [0, 1] \rightarrow \mathcal{R}$ from $\rho_S(1)$ to a pair $\rho_N(1)$ of matrices whose spectra lie on the unit circle. As these matrices preserve orientation, the dimensions of the maximal subspaces $V_{--}, V_{-+},$ and V_{+-} of $\mathbb{R}^n$ on which they act as $(-I, -I), (-I, I),$ and

$(I, -I)$ must then have the same parity $\nu \in \{0, 1\}$. By semisimplicity, there exists $B \in \mathrm{GL}^+(n)$ such that conjugating these matrices by B gives

$$(3) \quad r'_k \oplus A_1 \oplus \cdots \oplus A_\mu \oplus I_{n-3\nu-2\mu}, \quad r'_i \oplus C_1 \oplus \cdots \oplus C_\mu \oplus I_{n-3\nu-2\mu},$$

where r'_k, r'_i are r_k, r_i respectively when $\nu = 1$ and are the 0×0 matrix otherwise and where we have $A_l, C_l \in \mathrm{SO}(2)$ for each l . Conjugating the constant path $[0, 1] \rightarrow \{\rho_N(1)\}$ by a smooth path $[0, 1] \rightarrow \mathrm{GL}^+(n)$ from the identity to B , we then get a smooth path $\rho_O : [0, 1] \rightarrow \mathcal{R}$ from $\rho_N(1)$ to the pair (3). A final smooth path $\rho_R : [0, 1] \rightarrow \mathcal{R}$ from $\rho_O(1)$ to $(r'_k \oplus I_{n-3}, r'_i \oplus I_{n-3})$ can then be defined by simultaneously rotating A_l and C_l within $\mathrm{SO}(2)$ to I_2 for each l . Concatenating $\rho_S, \rho_N, \rho_O, \rho_R$, smoothing, and reparametrizing to the interval $[0, 1]$, we get Remark 2.3:

Remark 2.3. For any integer $n \geq 3$ and commuting $A, C \in \mathrm{GL}^+(n)$, we may construct smooth paths

$$\alpha, \gamma : [0, 1] \rightarrow \mathrm{GL}^+(n)$$

such that $[\alpha, \gamma]$ is identically the identity and

$$\alpha(0) = A, \quad \gamma(0) = C, \quad \alpha(1) = r'_k \oplus I_{n-3}, \quad \gamma(1) = r'_i \oplus I_{n-3},$$

where r_k, r_i are as in (1) and $\nu \in \{0, 1\}$ is the parity of the dimension of the maximal subspace of $\mathbb{R}^n$ on which all generalized eigenvalues of A and C are negative real numbers.

2.2. Local standardization. For $a, c \in \mathrm{Diff}^+(X)$ and $x \in X$ as in Proposition 2.1 (so, note we must have $n \geq 3$), we now produce diffeomorphisms $a', c' \in \mathrm{Diff}^+(X)$ whose commutator $[a', c']$ is smoothly isotopic to the identity through diffeomorphisms that each fix a neighborhood of x such that, for r_k and r_i as in (1), a' and c' act as $r_k \oplus I_{n-3}$ and $r_i \oplus I_{n-3}$ respectively in some local coordinates about x . We then discuss the consequence of this result that the connected sum $X_1 \# X_2$ of any manifolds X_1, X_2 of the same dimension at least 4 each admitting diffeomorphisms as in Proposition 2.1 will, in addition, admit such diffeomorphisms. To begin, fixing a smooth function $\chi : [0, \infty) \rightarrow [0, 1]$ which evaluates to 1 on $[0, 1]$ and to 0 on $[2, \infty)$ and local coordinates $\psi : U \rightarrow \mathbb{R}^n$ on X taking x to the origin, we take $f \in \mathrm{Diff}^+(X)$ fixing x such that $f \circ \psi^{-1}$ maps the ball B_3 of radius 3 in $\mathbb{R}^n$ into U (so, $f^\psi := \psi \circ f \circ \psi^{-1}|_{B_3}$ is a smooth embedding $B_3 \rightarrow \mathbb{R}^n$). For a smooth path $\rho : [0, 1] \rightarrow \mathrm{GL}^+(n)$, we consider the family

$$B_3 \rightarrow TB_3, \quad v \mapsto \chi(|v|) \frac{d\rho}{dt}(t) \rho(t)^{-1} v \in T_v B_3 \quad (t \in [0, 1])$$

of vector fields on B_3 and its flow $\Psi_t^\rho : B_3 \rightarrow B_3$, so $f_t^{\psi, \rho} := f^\psi \circ \Psi_t^\rho$ is a smooth embedding $B_3 \rightarrow \mathbb{R}^n$ which agrees with f^ψ on $B_3 \setminus B_2$ and satisfies $d(f_t^{\psi, \rho})_0 = df_0^\psi \circ \rho(t)$ for each $t \in [0, 1]$. For small enough $\varepsilon \in (0, 1)$,

$$f_t^{\psi, \varepsilon} : B_3 \rightarrow \mathbb{R}^n, \quad v \mapsto f^\psi(v) - t\chi(|v|/\varepsilon)(f^\psi(v) - d(f^\psi)_0(v))$$

is also a smooth embedding which agrees with f^ψ on $B_3 \setminus B_2$ and satisfies $d(f_t^{\psi,\varepsilon})_0 = df_0^\psi$ for each $t \in [0, 1]$, and which agrees with df_0^ψ on B_ε when $t = 1$. We then respectively extend

$$\psi^{-1} \circ f_t^{\psi,\rho} \circ \psi|_{\psi^{-1}(B_3)}, \quad \psi^{-1} \circ f_t^{\psi,\varepsilon} \circ \psi|_{\psi^{-1}(B_3)}$$

to diffeomorphisms $f_t^\rho, f_t^\varepsilon \in \text{Diff}^+(X)$ (each smoothly varying in t) by setting them equal to f on $X \setminus \psi^{-1}(B_3)$ for each $t \in [0, 1]$.

Potentially rescaling the chosen local coordinates ψ , we may arrange that $a \circ \psi^{-1}$ and $c \circ \psi^{-1}$ map B_3 into U . Considering $A := da_0^\psi$ and $C := dc_0^\psi$ alongside α and γ as in Remark 2.3, we then get smooth isotopies $a_t^{A^{-1}\alpha}$ and $c_t^{C^{-1}\gamma}$ through diffeomorphisms whose differentials at x commute for all $t \in [0, 1]$ that respectively take a and c to diffeomorphisms $a^\sim := a_1^{A^{-1}\alpha}$ and $c^\sim := c_1^{C^{-1}\gamma}$ which satisfy $d(a^\sim)_0^\psi = r_k \oplus I_{n-3}$ and $d(c^\sim)_0^\psi = r_i \oplus I_{n-3}$. For small enough $\varepsilon > 0$, we then get further smooth isotopies $(a^\sim)_t^\varepsilon$ and $(c^\sim)_t^\varepsilon$ through diffeomorphisms whose differentials at x commute for each $t \in [0, 1]$ that respectively take $a^\sim$ and $c^\sim$ to diffeomorphisms $a' := (a^\sim)_1^\varepsilon$ and $c' := (c^\sim)_1^\varepsilon$ which satisfy

$$(4) \quad (a')^\psi|_{B_\varepsilon} = r_k \oplus I_{n-3}|_{B_\varepsilon}, \quad (c')^\psi|_{B_\varepsilon} = r_i \oplus I_{n-3}|_{B_\varepsilon}.$$

We may then pair these smooth isotopies with our assumption that $[a, c]$ is smoothly isotopic to the identity through diffeomorphisms whose differentials at x each act as the identity to show existence of a smooth isotopy b_t from $b_0 = [a', c']$ to $b_1 = \text{id}$ through such diffeomorphisms. Appropriately rescaling ψ (such that (4) still holds), we may also arrange that each $b_t \circ \psi^{-1}$ maps B_3 into U . Taking $\delta \in (0, \varepsilon/2]$ small enough such that b_t^δ is a diffeomorphism for each t , we get a smooth isotopy taking $b_0^\delta = [a', c']$ to $b_1^\delta = \text{id}$ through diffeomorphisms that each fix $\psi^{-1}(B_\delta)$. Writing $O := \psi^{-1}(B_\delta)$ and appropriately rescaling ψ to produce new local coordinates $\phi : O \rightarrow \mathbb{R}^n$, we have shown Remark 2.4 (and, as a consequence, Remark 2.5):

Remark 2.4. For $a, c \in \text{Diff}^+(X)$ as in Proposition 2.1, we may produce diffeomorphisms $a', c' \in \text{Diff}^+(X)$ whose commutator $[a', c']$ is smoothly isotopic to the identity through diffeomorphisms that each fix a neighborhood O of x , on which we may pick local coordinates $\phi : O \rightarrow \mathbb{R}^n$ taking x to the origin such that, with r_k and r_i as in (1), we have

$$\phi \circ a' \circ \phi^{-1} = r_k \oplus I_{n-3}, \quad \phi \circ c' \circ \phi^{-1} = r_i \oplus I_{n-3}.$$

Remark 2.5. Consider smooth manifolds X_1, X_2 of the same dimension n alongside diffeomorphisms $a_i, c_i \in \text{Diff}^+(X_i)$ for $i \in \{1, 2\}$ as in Proposition 2.1. Picking $a'_i, c'_i \in \text{Diff}^+(X_i)$ and $\phi_i : O_i \rightarrow \mathbb{R}^n$ as in Remark 2.4 and taking the connected sum

$$X := X_1 \# X_2 := (X_1 \setminus \phi_1^{-1}(B_1)) \cup (X_2 \setminus \phi_2^{-1}(B_1)),$$

the resulting connected sums $a := a'_1 \# a'_2$ and $c := c'_1 \# c'_2$ are well-defined elements of $\text{Diff}^+(X)$. When $n \geq 4$, we can directly see that X , a , c , and $x := \phi_1^{-1}(0, 0, 0, 2, 0, \dots, 0) \in X$ satisfy the antecedent of Proposition 2.1.

We also note that it is straightforward to see a stronger statement for products: for any smooth manifolds X_1, X_2 and $a_1^\circ, c_1^\circ \in \text{Diff}^+(X_1^\circ, \partial)$ whose commutator $[a_1^\circ, c_1^\circ]$ represents the smooth mapping class rel boundary of the boundary Dehn twist on X_1° , writing $X := X_1 \times X_2$ and denoting by a_1, c_1 the extensions of a_1°, c_1° to X_1 , the diffeomorphisms $a_1 \times \text{id}, c_1 \times \text{id} \in \text{Diff}^+(X)$ restrict to diffeomorphisms $a^\circ, c^\circ \in \text{Diff}^+(X^\circ, \partial)$ whose commutator $[a^\circ, c^\circ]$ can be seen to represent the smooth mapping class rel boundary of the boundary Dehn twist on X° . We can therefore observe that, for X any manifold to which Theorem 1.1 applies, the theorem also holds for the product of X with any manifold.

2.3. Boundary Dehn twists as commutators. For $a, c \in \text{Diff}^+(X)$ as in Proposition 2.1, we now construct $a^\circ, c^\circ \in \text{Diff}^+(X^\circ, \partial)$ whose commutator $[a^\circ, c^\circ]$ represents the smooth mapping class rel boundary of the boundary Dehn twist on X° . To this end, we consider $a', c' \in \text{Diff}^+(X)$ and $\phi : O \rightarrow \mathbb{R}^n$ as in Remark 2.4 and write $X^\circ := X \setminus \phi^{-1}(B_1)$, where B_r is the ball of radius r about the origin in $\mathbb{R}^n$. Translating, extending, and smoothing the paths $R_k \oplus I_{n-3}$ and $R_i \oplus I_{n-3}$ as in (2), we produce smooth paths

$$\rho_k, \rho_i : [1, \infty) \rightarrow \text{SO}(n)$$

which both evaluate to the identity on $[1, 2]$ and respectively evaluate to $r_k \oplus I_{n-3}$ and $r_i \oplus I_{n-3}$ on $[3, \infty)$. We then consider diffeomorphisms a° and c° of $X^\circ \ni x'$ defined by

$$a^\circ(x') := (\phi^{-1} \circ \rho_k(|\phi(x')|) \circ \phi)(x'), \quad c^\circ(x') := (\phi^{-1} \circ \rho_i(|\phi(x')|) \circ \phi)(x')$$

when $x' \in O \setminus \phi^{-1}(B_1)$ and

$$a^\circ(x') := a'(x'), \quad c^\circ(x') := c'(x')$$

otherwise. a° and c° will then be orientation-preserving and will restrict to the identity on $\phi^{-1}(B_2 \setminus B_1)$, so we have $a^\circ, c^\circ \in \text{Diff}^+(X^\circ, \partial)$.

Lemma 2.6. *For $a, c \in \text{Diff}^+(X)$ as in Proposition 2.1, the commutator $[a^\circ, c^\circ]$ of the diffeomorphisms $a^\circ, c^\circ \in \text{Diff}^+(X^\circ, \partial)$ constructed from a, c represents the smooth mapping class rel boundary of the boundary Dehn twist on X° .*

Proof. Writing $\nu(\partial X^\circ) := O \setminus \phi^{-1}(B_1)$, the commutator $[a^\circ, c^\circ]|_{X \setminus \nu(\partial X^\circ)} = [a', c']|_{X \setminus O}$ is smoothly isotopic to the identity rel boundary on $X \setminus \nu(\partial X^\circ)$, so $[a^\circ, c^\circ]$ is smoothly isotopic rel boundary to the diffeomorphism of X° which equals $[a^\circ, c^\circ]$ on $\nu(\partial X^\circ)$ and the identity elsewhere. We may therefore complete the proof by showing that this diffeomorphism represents the

smooth mapping class rel boundary of the boundary Dehn twist on X° . To this end, considering the map

$$\varphi : \nu(\partial X^\circ) \rightarrow [1, \infty) \times S^{n-1}, \quad \phi^{-1}(v) \mapsto (|v|, v/|v|),$$

we can observe that

$$\varphi \circ [a^\circ, c^\circ] \circ \varphi^{-1} : (t, \omega) \mapsto (t, [\rho_k(t), \rho_i(t)]\omega).$$

Remark 2.2 then shows that the homotopy class of the loop $[\rho_k, \rho_i]$ will generate $\pi_1(SO(n))$. Therefore, we can see from the definition of the boundary Dehn twist presented in Section 1 that the smooth mapping class rel boundary of the diffeomorphism of X° which equals $[a^\circ, c^\circ]$ on $\nu(\partial X^\circ)$ and the identity elsewhere is exactly that of the boundary Dehn twist on X° . As we also justified why this diffeomorphism is isotopic rel boundary to the commutator $[a^\circ, c^\circ]$ of a° and c° , the proof is complete. $\square$

Consider the diffeomorphisms $a^\sim := a^\circ, c^\sim := c^\circ$ of $D^4 \cong (S^4)^\circ$ obtained by applying the above construction with a and c the maps

$$\omega \mapsto (-\omega_0, -\omega_1, \omega_2, \omega_3, \omega_4), \quad \omega \mapsto (\omega_0, -\omega_1, \omega_2, -\omega_3, \omega_4)$$

on $X = S^4$ and $x = (0, 0, 0, 0, 1)$. Now taking notation as in Proposition 2.1 and a', c' as in Remark 2.4, we may (as in Remark 2.5) take the connected sum $D^4 \# X$ such that the resulting connected sums $a^\sim \# a'$ and $c^\sim \# c'$ are well-defined elements of $\text{Diff}^+(D^4 \# X, \partial)$ which are respectively smoothly isotopic rel boundary to a° and c° under an identification $D^4 \# X \cong X^\circ$. It may be of interest to observe that Baraglia and Konno, in formalizing their elegant proof that the boundary Dehn twist is nontrivial in the smooth mapping class group rel boundary in many cases [5, Theorem 1.5], made use of certain diffeomorphisms σ'_1, σ'_2 of D^4 [5, Subsections 4.1-4.2], to which we can observe $a^\sim$ and $c^\sim$ are respectively smoothly isotopic rel boundary. In their use of these diffeomorphisms, they extend them to diffeomorphisms σ_1^X, σ_2^X of an arbitrary 4-manifold $X \supset D^4$ by setting them equal to the identity on $X \setminus D^4$. To avoid confusion, we note that the point here is different than in our setting: σ_1^X and σ_2^X do not then give rise to a commutator representative for the boundary Dehn twist on X° in the smooth mapping class group rel boundary (except in the trivial case when this boundary Dehn twist is trivial in the smooth mapping class group rel boundary). However, σ_1^X and σ_2^X do admit a shared fixed point x as in Proposition 2.1 and have commutator that is smoothly isotopic to the identity, though not through diffeomorphisms whose differentials at x each act as the identity, so they give examples of why this condition is necessary in the proposition.

2.4. Non-spin mapping tori. Consider the standard CW decomposition

$$T^2 = e^0 \cup e_a^1 \cup e_c^1 \cup e^2$$

of the 2-torus with one 0-cell e^0 , two 1-cells e_a^1, e_c^1 , and one 2-cell e^2 . For diffeomorphisms a and c of a smooth manifold X , we may first consider the

mapping tori $X_a \rightarrow e^0 \cup e_a^1$ and $X_c \rightarrow e^0 \cup e_c^1$, the wedge of which gives an X -family over the 1-skeleton $e^0 \cup e_a^1 \cup e_c^1$ of T^2 . A smooth isotopy from the commutator $[a, c]$ of a and c to the identity then exactly directs how to extend this family from the 1-skeleton to the 2-cell, giving rise to the mapping torus $X_{a,c} \rightarrow T^2$ of a and c , a smooth X -bundle over T^2 . When a and c are orientation-preserving, this bundle will also be orientable. Thus, we may complete the proof of the proposition by verifying Lemma 2.7:

Lemma 2.7. *For $a, c \in \text{Diff}^+(X)$ as in Proposition 2.1, the total space $X_{a,c}$ of the mapping torus of a and c is not spin.*

Proof. By assumption, we may pick a smooth isotopy from the commutator $[a, c]$ to the identity through diffeomorphisms whose differentials at x each act as the identity; we construct the mapping torus $X_{a,c} \rightarrow T^2$ using this isotopy as described above. Denoting by $(T_x X)_{da_x, dc_x} \rightarrow T^2$ the mapping torus of da_x and dc_x , this construction makes clear that for $s : T^2 \rightarrow X_{a,c}$ the section of the bundle $X_{a,c} \rightarrow T^2$ taking $z \mapsto (z, x)$, we have

$$s^*T(X_{a,c}) \cong TT^2 \oplus (T_x X)_{da_x, dc_x}.$$

Fixing any spin structure on T^2 , any spin structure on $X_{a,c}$ will therefore give rise to a spin structure on $(T_x X)_{da_x, dc_x}$. Now, taking local coordinates $\psi : U \rightarrow \mathbb{R}^n$ on X taking x to the origin, we write $A := d(\psi \circ a \circ \psi^{-1})_0$ and $C := d(\psi \circ c \circ \psi^{-1})_0$ to enter alignment with the notation of Remark 2.3, showing that there exist smooth paths through commuting matrices taking A and C to the matrices $r_k \oplus I_{n-3}$ and $r_i \oplus I_{n-3}$ respectively as in (1). We then get a bundle isomorphism

$$\mathbb{R}_{r_k \oplus I_{n-3}, r_i \oplus I_{n-3}}^n \cong (T_x X)_{da_x, dc_x},$$

where the left-hand side of this isomorphism is the mapping torus of $r_k \oplus I_{n-3}$ and $r_i \oplus I_{n-3}$. This mapping torus will then be spin exactly when the mapping torus $\mathbb{R}_{r_k, r_i}^3 \rightarrow T^2$ of r_k and r_i is spin. However, the definition (1) of r_k and r_i directly shows that $\mathbb{R}_{r_k, r_i}^3$ is a familiar non-spin bundle over T^2 (as can be formally verified using Remark 2.2), so $(T_x X)_{da_x, dc_x}$ —and thus $X_{a,c}$ —can also not be spin. $\square$

2.5. Commutators in other groups. Consider an oriented n -manifold X and the group $\text{Diff}(X)$ of all diffeomorphisms of X . In analogy with the summary of results presented in Section 1, we can see that X admits a pair $a, c \in \text{Diff}(X)$ of commuting involutions (which respectively preserve and reverse orientation) with a shared fixed point $x \in X$ that respectively act as the antipodal map in one complex coordinate and complex conjugation in suitable local coordinates about x for X any complete intersection of odd complex dimension (in a complex projective space) by Proposition 3.1, any complete intersection of odd complex dimension cut out in a product of complex projective spaces by polynomials whose multidegrees satisfy a certain

constraint by Proposition 4.1, or any holomorphic double branched cover of odd complex dimension with compatible real structure whose branch divisor contains a real point. For an orientation-reversing involution $\tau \in O(n)$, we denote by $\text{Diff}^\tau(X^\circ, \partial)$ the group of diffeomorphisms of X° which restrict to either the identity or $\text{id} \times \tau$ on some neighborhood $\nu(\partial X^\circ) \cong [0, 1] \times S^{n-1}$ of ∂X° . An analogue of Proposition 2.1 in this setting then gives a construction of $a^\circ, c^\circ \in \text{Diff}^\tau(X^\circ, \partial)$ whose commutator $[a^\circ, c^\circ] \in \text{Diff}^+(X^\circ, \partial)$ represents the smooth mapping class rel boundary of the boundary Dehn twist on X° for all such X (and similarly gives that the mapping torus $X_{a,c}$ of a, c is a smooth nonorientable X -bundle over T^2 whose total space is not Pin^+). Then, for c' an orientation-reversing diffeomorphism of any manifold X' admitting a fixed point, $a \times \text{id}$ and $c \times c'$ are orientation-preserving diffeomorphisms of $X \times X'$ as in Proposition 2.1, so we get a concrete realization of the smooth mapping class rel boundary of the boundary Dehn twist on $(X \times X')^\circ$ as a commutator in $\text{Diff}^+((X \times X')^\circ, \partial)$.

Moving outside of the smooth category, we consider a symplectic manifold (X, ω) . For symplectomorphisms a, c of X with shared fixed point $x \in X$, the maximal subspace of $T_x X$ on which all generalized eigenvalues of the differentials da_x and dc_x are negative real numbers will be symplectic, and will thus have even dimension. Therefore, we cannot apply a direct analogue of Proposition 2.1 to produce a commutator in the symplectic mapping class group rel boundary which represents the symplectic mapping class rel boundary of the boundary Dehn twist, motivating further study of such objects using other approaches.

3. COMPLETE INTERSECTIONS

In this section, we prove Proposition 3.1, which shows that smooth complete intersections (and thus, as is discussed in Remark 2.5, all manifolds resulting from connected sums thereof) satisfy the antecedent of Proposition 2.1, verifying that Theorem 1.1 holds for all such spaces.

Proposition 3.1. *Any smooth complete intersection X admits commuting smooth involutions a and c with a shared fixed point x which respectively act as the antipodal map in one complex coordinate and complex conjugation in suitable complex local coordinates about x (so, a is always orientation-preserving, while c is orientation-preserving exactly when X has even complex dimension).*

We first introduce complete intersections (by which we specifically mean positive-dimensional complete intersections in complex projective spaces) in Subsection 3.1, then justify why smooth complete intersections admit certain especially symmetric presentations in Subsection 3.2, and finally use the symmetries of these presentations to define the involutions whose existence Proposition 3.1 communicates in Subsection 3.3.

3.1. Definition and examples. Consider the n -(complex)-dimensional complex projective space $\mathbb{P}^n$ alongside $m < n$ homogeneous polynomials

$$p := (p_1, \dots, p_m) : \mathbb{C}^{n+1} \rightarrow \mathbb{C}^m$$

of degrees $d := (d_1, \dots, d_m)$, so we have

$$p_i(\lambda z) = \lambda^{d_i} p_i(z)$$

for any $z \in \mathbb{C}^{n+1}$ and $\lambda \in \mathbb{C}$. The set

$$X := \{[z] \in \mathbb{P}^n \mid p(z) = 0\}$$

is well-defined and called *the subset of $\mathbb{P}^n$ cut out by p* . We say a point $x \in X$ is *singular* when the differential of the map from homogeneous coordinates on $\mathbb{P}^n$ about x to $\mathbb{C}^m$ induced by p fails to be surjective at x and that X is a *smooth complete intersection* when X has no singular points, in which case the implicit function theorem shows X will be a complex manifold of dimension $n - m$. We assume that any subset X of $\mathbb{P}^n$ referred to as a smooth complete intersection has positive dimension $n - m \geq 1$. The Lefschetz hyperplane theorem [29, Theorem 7.4] gives that any such X will be connected and, when $n - m \geq 2$, simply-connected. Moreover, by an observation of Thom, a generic choice of homogeneous polynomials of degrees d will cut out a smooth complete intersection in $\mathbb{P}^n$ diffeomorphic to X .

For any positive integers n and d , we can see that $z_0^d + \dots + z_{n-1}^d + z_{n-1}z_n^{d-1}$ cuts out a smooth complex hypersurface in $\mathbb{P}^n$, which we denote by X_d when $n = 3$. We then have that the underlying smooth manifold of X_2 is $S^2 \times S^2$, and of X_4 is the $K3$ surface. In addition, the infinite family of smooth manifolds underlying the complex hypersurfaces X_{32m+4} for m an integer congruent to 0 or 1 modulo 3 are among those whose punctures' boundary Dehn twists Baraglia and Konno showed [5, Theorem 1.5] are nontrivial in the smooth mapping class group rel boundary, as we can see [11, Exercises 1.3.13(c)(e)] that a smooth complete intersection X cut out by $n - 2$ polynomials of degrees $d_1, \dots, d_{n-2}$ in $\mathbb{P}^n$ has first Chern class $d_1 + \dots + d_{n-2} - n - 1$ times a primitive class and signature

$$\frac{1}{3}(n + 1 - d_1^2 - \dots - d_{n-2}^2)d_1 \cdots d_{n-2}.$$

The polynomials cutting out these complex hypersurfaces exhibit symmetries (for d even) which we show can be much more broadly realized in Subsection 3.2.

3.2. Symmetric homogeneous cutting polynomials. We now prove Lemma 3.2, which shows that any smooth complete intersection is diffeomorphic to a complete intersection cut out by polynomials exhibiting certain symmetries used in Lemma 3.3 to define the involutions whose existence Proposition 3.1 communicates.

Lemma 3.2. *For any positive integer n and finite tuple d of positive integers, consider a smooth complete intersection X cut out in $\mathbb{P}^n$ by polynomials of degrees d . A generic choice of homogeneous polynomials of degrees d with only real coefficients whose constituent monomials each have even power of z_0 and positive power of some coordinate function other than z_n will cut out a complete intersection in $\mathbb{P}^n$ diffeomorphic to X .*

Proof. With notation as in the lemma statement, we write $d = (d_1, \dots, d_m)$, so d_i is the degree of the i th polynomial cutting out X and $m < n$, and set $Q := Q_1 \times \dots \times Q_m$ for Q_i the complex vector space of homogeneous polynomials of degree d_i on $\mathbb{C}^{n+1}$ whose constituent monomials each have even power of z_0 and positive power of some coordinate function other than z_n . For a subset S of $\mathbb{P}^n$ and Δ the map taking an $m \times (n+1)$ matrix to the $\binom{n+1}{m}$ -tuple of determinants of its $m \times m$ minors, we denote by

$$\mathcal{D}(S) := \{(q, [s]) \in Q \times S \mid (q(s), \Delta(dq_s)) = 0\}$$

the subset of $Q \times S$ cut out by the homogeneous polynomials

$$(q, z) \mapsto (q(z), \Delta(dq_z))$$

on $Q \times \mathbb{C}^{n+1} \ni (q, z)$, so $(q, [s]) \in \mathcal{D}(S)$ exactly when $[s] \in S$ is a singular point of the subset of $\mathbb{P}^n$ cut out by $q \in Q$. Considering the projection $\mathcal{Q} : Q \times \mathbb{P}^n \rightarrow Q$, the set $D(S) := \mathcal{Q}(\mathcal{D}(S))$ is then exactly the set of $q \in Q$ which cut out a subset of $\mathbb{P}^n$ which has singular points in S . When S is Zariski-closed in $\mathbb{P}^n$, $Q \times S$ will be Zariski-closed in $Q \times \mathbb{P}^n$, so $\mathcal{D}(S)$ will be Zariski-closed in $Q \times \mathbb{P}^n$. $\mathcal{Q}$ is proper, so $D(S)$ will be Zariski-closed in the complex vector space Q , and will thus be cut out in Q (whose ring of polynomials we note is Noetherian by Hilbert's basis theorem [2, Theorem 7.5]) by finitely many polynomials. Then, writing $D := D(\mathbb{P}^n)$ and denoting by R the real vector space of elements of Q with only real coefficients, $D \cap R$ is cut out in R by the real and imaginary parts of the finitely many polynomials which cut out D in Q . Any polynomials which vanish on the entirety of R must vanish on the entirety of Q , meaning that $D \cap R$ will be a proper subset of R if D is a proper subset of Q . In this case, because $D \cap R$ will be cut out by finitely many polynomials which do not all vanish on all of R , a generic choice of $r \in R$ will lie outside of D . Recalling that $r \in R$ lies outside of D exactly when r cuts out a smooth complete intersection in $\mathbb{P}^n$ and that such a smooth complete intersection will be diffeomorphic to X by the observation of Thom that any two smooth complete intersections cut out in $\mathbb{P}^n$ by polynomials of the same degrees are diffeomorphic (in fact, even smoothly ambiently isotopic; this follows from an argument similar to that which we are now exhibiting), the proof will be complete if we show that D is a proper subset of Q . Noting that a union of two proper Zariski-closed subsets of Q will certainly still be a proper subset of Q , we set

$$A := \{[a] \in \mathbb{P}^n \mid a_1 = \dots = a_{n-1} = 0\}, \quad B := \mathbb{P}^n \setminus A,$$

so we have $D = D(A) \cup D(B)$. We may thus complete the proof by showing that the Zariski-closures of $D(A)$ and $D(B)$ are proper subsets of Q .

As A is a Zariski-closed subset of $\mathbb{P}^n$, the argument presented above shows that $D(A)$ will be a Zariski-closed subset of Q . Therefore, we need only show that $D(A)$ is a proper subset of Q . Writing $q_A := (q_{A,1}, \dots, q_{A,m})$ for

$$q_{A,i}(z) := \begin{cases} z_i, & d_i = 1 \\ z_0^{d_i-1} z_i + z_0^{d_i-1} z_n + z_i z_n^{d_i-1} & d_i > 1 \text{ odd}, \\ z_0^{d_i} + z_i z_n^{d_i-1}, & d_i \text{ even} \end{cases}$$

we can see that $q_{A,i} \in Q_i$ (where, when d_i is odd, we note that in light of the standing assumption that smooth complete intersections are positive-dimensional we initiated when presenting their definition, we have $m < n$), so $q_A \in Q$. Additionally, denoting by $\delta_{i,j}$ the Kronecker delta (which equals 1 when $i = j$ and 0 otherwise), we can then directly compute in each case that at any point $[a] \in A$ such that $q_{A,i}(a) = 0$, we have

$$\left[\frac{\partial q_{A,i}}{\partial z_1} \quad \dots \quad \frac{\partial q_{A,i}}{\partial z_m} \right] (a) = [\delta_{i,1} \quad \dots \quad \delta_{i,m}].$$

Therefore, for any $[a] \in A$ at least one of the conditions

$$q_A(a) \neq 0, \quad \Delta(d(q_A)_a) \ni \det \left[\frac{\partial q_A}{\partial z_1} \quad \dots \quad \frac{\partial q_A}{\partial z_m} \right] (a) = 1 \neq 0$$

is satisfied, so $q_A \notin D(A)$. This completes our proof that the Zariski-closure of $D(A)$ (which coincides with $D(A)$) is a proper subset of Q .

We now complete the proof by showing that the Zariski-closure of $D(B)$ is a proper subset of Q . To this end, pick $[b] \in B$, so there exists $j \in \{1, \dots, n-1\}$ such that $b_j \neq 0$. Then, for

$$q_{b,i}(z) := (\delta_{i,1} z_j^{d_1}, \dots, \delta_{i,m} z_j^{d_m}) \in Q,$$

we see that the set $\{q_{b,1}(b), \dots, q_{b,m}(b)\}$ spans $\mathbb{C}^m$. Therefore, the kernel of the linear evaluation map

$$\text{ev}_{[b]} : Q \rightarrow \mathbb{C}^m, \quad q \mapsto q(b/b_j)$$

has dimension $\dim Q - m$. Denoting by B_j the set of all $[b] \in B$ satisfying $b_j \neq 0$, we then have a morphism

$$Q \times B_j \rightarrow \mathbb{C}^m \times B_j, \quad (q, [b]) \mapsto (q(b/b_j), [b])$$

of (trivial) algebraic vector bundles over the Zariski-open subset B_j of $\mathbb{P}^n$. We then consider the kernel bundles $\mathcal{E}_j \rightarrow B_j$ of these morphisms for $j \in \{1, \dots, n-1\}$, which we can observe give rise to an algebraic vector subbundle $\mathcal{E} \rightarrow B$ of the trivial bundle $Q \times B \rightarrow B$ over the Zariski-open subset B of $\mathbb{P}^n$. Notably, we may conclude that the total space $\mathcal{E}$ of this bundle is a smooth subvariety of $Q \times B$, and thus that the projection $\mathcal{Q}|_{\mathcal{E}} : \mathcal{E} \rightarrow \overline{E}$ is a morphism of varieties over $\mathbb{C}$ with nonsingular domain (in the language of the book of Hartshorne [13, Remark II.4.10.1]), where

$\overline{E}$ is the Zariski-closure of $E := \mathcal{Q}(\mathcal{E})$ in Q . Therefore, we can apply the generic smoothness theorem [13, Corollary III.10.7] to see that $\mathcal{Q}|_{\mathcal{E}}$ will be smooth on $\mathcal{Q}|_{\mathcal{E}}^{-1}(\overline{E} \setminus F)$ for a proper Zariski-closed subset F of $\overline{E}$. As the fiber $\mathcal{Q}|_{\mathcal{E}}^{-1}(q)$ is exactly $\{q\}$ times the intersection of the subset of $\mathbb{P}^n$ cut out by q with B , $D(B)$ lies in the proper Zariski-closed subset F of Q , so the Zariski-closure of $D(B)$ is a proper subset of Q . In light of our above proof that the Zariski-closure of $D(A)$ is a proper subset of Q and the fact that $D = D(A) \cup D(B)$, this completes the proof that D is a proper subset of Q , which we observed above completes the proof altogether. $\square$

3.3. Involutions of complete intersections. Lemma 3.3 states that any smooth complete intersection cut out by polynomials exhibiting the symmetries of interest in Lemma 3.2 admits involutions as in Proposition 3.1:

Lemma 3.3. *For any positive integer n and any smooth complete intersection X cut out in $\mathbb{P}^n$ by polynomials with only real coefficients whose constituent monomials each have even power of z_0 and positive power of some coordinate function other than z_n , the maps*

$$(5) \quad [z] \mapsto [-z_0 : z_1 : \cdots : z_n], \quad [z] \mapsto [\overline{z}]$$

on $\mathbb{P}^n$ respectively restrict to commuting smooth involutions a and c of X which act as the antipodal map in one complex coordinate and complex conjugation in suitable complex local coordinates about their shared fixed point $[0 : \cdots : 0 : 1] \in X$.

Proof. With notation as in the lemma statement, say X is cut out by m polynomials p . With $x := [0 : \cdots : 0 : 1] \in \mathbb{P}^n$, we may observe that $p(x) = 0$, so $x \in X$. We may consider complex coordinates

$$\varphi : z \mapsto [z_0/z_n : \cdots : z_{n-1}/z_n : 1]$$

in a neighborhood of x on $\mathbb{P}^n$, and the differential of

$$p_\varphi : z \mapsto p(z_0/z_n, \dots, z_{n-1}/z_n, 1),$$

at $x \in X$ is then surjective because X is smooth. Note also that $\partial_0 p_\varphi(x) = 0$, as $x_0 = 0$ and the polynomials p_φ admit decompositions into summand monomials whose powers of z_0 are all even (and are, notably, thus not 1). From these two facts, we may consider a permutation $\sigma : \{0, \dots, n-1\} \rightarrow \{0, \dots, n-1\}$ such that the vectors $((\partial_{\sigma(i)} p_\varphi)(x))_{i=n-m}^{n-1}$ are linearly independent and $\sigma(0) = 0$. Therefore, the implicit function theorem states that there exists a unique smooth function P_φ such that

$$(p_\varphi \circ \sigma)(z_{\sigma(0)}, \dots, z_{\sigma(n-m-1)}, P_\varphi(z_{\sigma(0)}, \dots, z_{\sigma(n-m-1)})) = 0,$$

giving complex coordinates

$$\varphi X : (z_0, \dots, z_{n-m-1}) \mapsto \varphi(z_{\sigma(0)}, \dots, z_{\sigma(n-m-1)}, P_\varphi(z_{\sigma(0)}, \dots, z_{\sigma(n-m-1)}))$$

about x in X . Now, the commuting smooth involutions (5) restrict to commuting smooth involutions a and c of X because the polynomials p cutting

out X admit decompositions into summand monomials whose powers of z_0 are all even and because the polynomials have only real coefficients. Then,

$$\begin{aligned} (\varphi_X^{-1} \circ a \circ \varphi_X)(z_0, \dots, z_{n-m-1}) &= (-z_0, z_1, \dots, z_{n-m-1}), \\ (\varphi_X^{-1} \circ c \circ \varphi_X)(z_0, \dots, z_{n-m-1}) &= (\bar{z}_0, \dots, \bar{z}_{n-m-1}), \end{aligned}$$

exactly as desired. $\square$

With X as in Lemma 3.3, observe that the fixed points of a are exactly

$$X \cap (\{[z] \in \mathbb{P}^n \mid z_0 = 0\} \cup \{[1 : 0 : \dots : 0]\}),$$

so when $\dim_{\mathbb{C}} X \geq 2$, the quotient map taking X to X/a will be a holomorphic double branched cover with deck transformation a having compatible real structure c (whose branch divisor then contains the real point $[0 : \dots : 0 : 1]$) exactly when $[1 : 0 : \dots : 0] \notin X$. Noting that we will have $[1 : 0 : \dots : 0] \notin X$ exactly when any of the polynomials cutting out X has nonzero coefficient of the monomial with positive power of only z_0 , we will necessarily have $[1 : 0 : \dots : 0] \in X$ —and thus that X/a is singular—whenever all degrees of the polynomials cutting out X are odd. Otherwise, a generic choice of polynomials as in Lemma 3.2 will cut out a smooth complete intersection in $\mathbb{P}^n$ which does not contain $[1 : 0 : \dots : 0]$; this complete intersection can then be given the structure of a holomorphic double branched cover with compatible real structure whose branch divisor contains a real point.

As was mentioned in Subsection 3.1, the polynomials described in that subsection are all of the form of interest in Lemma 3.3, so we get explicit descriptions of how involutions of the complete intersections they cut out which satisfy the conditions of interest in the lemma arise. For example, in the case that $X = X_2$, these smooth involutions (under some identification $X_2 \cong S^2 \times S^2 \subset (\mathbb{C} \times \mathbb{R})^2$) take

$$(\eta_0, \xi_0, \eta_1, \xi_1) \mapsto (-\eta_0, \xi_0, \eta_1, \xi_1), \quad (\eta_0, \xi_0, \eta_1, \xi_1) \mapsto (\bar{\eta}_0, \xi_0, \bar{\eta}_1, \xi_1).$$

We also realize the smooth involutions produced on the $K3$ surface X_4 from another perspective which may be of interest; we formalize this and a generalization thereof which covers the setting of the simply-connected elliptic surfaces $E(n)$ in Subsection 4.3.

In addition to exhibiting involutions as in Lemma 3.3 on the smooth complete intersections described in Subsection 3.1, Lemma 3.3 can be combined with Lemma 3.2 to directly give the result of Proposition 3.1, whose proof was the main purpose of this section.

4. COMPLETE INTERSECTIONS IN PRODUCTS OF COMPLEX PROJECTIVE SPACES

In this section, we present Proposition 4.1, which shows that all smooth complete intersections cut out in products

$$\mathbb{P}^{\mathbf{n}} := \mathbb{P}^{n_0} \times \cdots \times \mathbb{P}^{n_\nu} \quad (\mathbf{n} := (n_0, \dots, n_\nu))$$

of complex projective spaces by polynomials whose multidegrees satisfy a certain constraint (and thus, as is discussed in Remark 2.5, all manifolds resulting from connected sums thereof) satisfy the antecedent of Proposition 2.1, verifying that Theorem 1.1 holds for all such spaces.

Proposition 4.1. *Consider a tuple $\mathbf{n} := (n_0, \dots, n_\nu)$ of positive integers, a positive integer $m < n := n_0 + \cdots + n_\nu$, and a $m \times (\nu + 1)$ matrix d of positive integers satisfying*

$$(6) \quad m_i = 0 \quad \text{or} \quad m - 2m_i \geq n - 2n_i$$

and $m_i < n_i$ for some $i \in \{0, \dots, \nu\}$, where m_i is the number of odd entries of the i th column of d . Any smooth complete intersection X cut out in the product $\mathbb{P}^{\mathbf{n}}$ of complex projective spaces by polynomials of multidegrees d admits commuting smooth involutions a and c with a shared fixed point $x \in X$ which respectively act as the antipodal map in one complex coordinate and complex conjugation in suitable complex local coordinates about x (so, a is always orientation-preserving, while c is orientation-preserving exactly when X has even complex dimension).

We first introduce complete intersections in products of complex projective spaces in Subsection 4.1, then present the fact that these spaces admit certain especially symmetric presentations analogous to those described in Subsection 3.2 when they satisfy the constraint (6) in Subsection 4.2, and finally use the symmetries of these presentations to define the involutions analogous to those described in Subsection 3.3 whose existence Proposition 4.1 communicates in Subsection 4.3.

4.1. Definition and examples. Consider a tuple $\mathbf{n} := (n_0, \dots, n_\nu)$ alongside $m < n := n_0 + \cdots + n_\nu$ multihomogeneous polynomials

$$p := (p_1, \dots, p_m) : \mathbb{C}^{n_0+1} \times \cdots \times \mathbb{C}^{n_\nu+1} \rightarrow \mathbb{C}^m$$

of multidegrees d , so we have

$$p_i(\lambda_0 z_0, \dots, \lambda_\nu z_\nu) = \lambda_0^{d_{i,0}} \cdots \lambda_\nu^{d_{i,\nu}} p_i(z_0, \dots, z_\nu)$$

for all $z_j \in \mathbb{C}^{n_j+1}$ and $\lambda_j \in \mathbb{C}$. The set

$$X := \{([z_0], \dots, [z_\nu]) \in \mathbb{P}^{\mathbf{n}} \mid p(z_0, \dots, z_\nu) = 0\}$$

is well-defined and called *the subset of $\mathbb{P}^{\mathbf{n}}$ cut out by p* . We say a point $x \in X$ is *singular* when the differential of the map from multihomogeneous coordinates on $\mathbb{P}^{\mathbf{n}}$ about x to $\mathbb{C}^m$ induced by p fails to be surjective at x and

that X is a *smooth complete intersection in a product of complex projective spaces* when it has no singular points, in which case it will be a complex manifold of dimension $n - m$. We assume any subset X of $\mathbb{P}^n$ referred to as a smooth complete intersection in a product of complex projective spaces has positive dimension $n - m \geq 1$. We also restrict ourselves to the case when all multidegrees cutting out such X have no zero entries: in this setting, the Lefschetz hyperplane theorem [29, Theorem 7.4] gives that X will be connected and, when $n - m \geq 2$, simply-connected. Moreover, the observation of Thom applies as in Subsection 3.1 to show that if X is a smooth complete intersection cut out in $\mathbb{P}^n$ by polynomials of multidegrees d , a generic choice of multihomogeneous polynomials of multidegrees d will cut out a complete intersection in $\mathbb{P}^n$ diffeomorphic to X .

Denote by $X_{2,m,n}$ the complex hypersurface cut out by the polynomial

$$z_{0,0}^2 p_1(z_{1,0}, z_{1,1}, z_{2,0}, z_{2,1}) + z_{0,1}^2 p_2(z_{1,0}, z_{1,1}, z_{2,0}, z_{2,1})$$

in $\mathbb{P}^1 \times \mathbb{P}^1 \times \mathbb{P}^1$, for polynomials p_1 and p_2 on $\mathbb{C}^2 \times \mathbb{C}^2$ of multidegrees (m, n) which satisfy the generic condition that they cut out smooth, transverse complex hypersurfaces in $\mathbb{P}^1 \times \mathbb{P}^1$ (for example, taking

$$(7) \quad \begin{aligned} p_1(z_{1,0}, z_{1,1}, z_{2,0}, z_{2,1}) &:= z_{1,0}^m z_{2,0}^n + 2z_{1,1}^m z_{2,0}^n + 3z_{1,0}^m z_{2,1}^n + \pi z_{1,1}^m z_{2,1}^n, \\ p_2(z_{1,0}, z_{1,1}, z_{2,0}, z_{2,1}) &:= z_{1,0}^m z_{2,0}^n + z_{1,1}^m z_{2,0}^n + z_{1,0}^m z_{2,1}^n + z_{1,0} z_{1,1}^{m-1} z_{2,1}^n, \end{aligned}$$

this will be true). The smooth 4-manifolds $X(m, n)$ underlying these complex surfaces make up an important family including the unique (up to diffeomorphism) simply-connected relatively minimal elliptic surface $E(n)$ with no multiple fibers and holomorphic Euler characteristic n for $m = 2$ (and thus including the $K3$ surface for $m = n = 2$) [11, Theorem 7.3.3]. The projection $X_{2,m,n} \rightarrow \mathbb{P}^1 \times \mathbb{P}^1$ forgetting the first factor realizes $X_{2,m,n}$ as the minimal resolution of the singular double branched cover of $\mathbb{P}^1 \times \mathbb{P}^1$ with branch divisor B the union of the (smooth and transverse) complex curves cut out by p_1 and p_2 . Denoting by Σ_g the surface of genus g , $\iota := \iota_g$ the hyperelliptic involution of Σ_g (an involution with $2g + 2$ fixed points $\text{Fix}(\iota)$) as in Figure 1, and $\pi : \Sigma_g \rightarrow \mathbb{P}^1 \cong \Sigma_g / \iota$ the projection, B has the same bidegree $(2m, 2n)$ as the divisor

$$B' := (\pi(\text{Fix}(\iota_{m-1})) \times \mathbb{P}^1) \cup (\mathbb{P}^1 \times \pi(\text{Fix}(\iota_{n-1}))).$$

The divisors B and B' are each reduced and nodal, so work [1] of Atiyah shows that the minimal resolutions of the singular double branched covers branched along each divisor will be diffeomorphic to some double branched covers branched along smooth divisors of bidegree $(2m, 2n)$, which will in turn be diffeomorphic to one another by path-connectedness of the space of smooth divisors of a given bidegree (as can be seen using Bertini's theorem [13, Corollary III.10.9]) combined with Ehresmann's fibration theorem [7]. Thus, $X(m, n)$ can also be realized as the minimal resolution of the singular double branched cover of $\mathbb{P}^1 \times \mathbb{P}^1$ with branch divisor B' . We can then

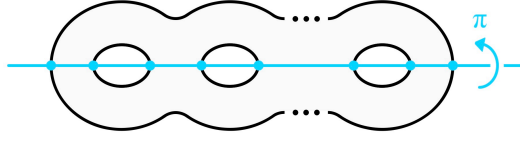


FIGURE 1. The hyperelliptic involution ι on the surface of genus g , with its $2g + 2$ fixed points marked. This involution is central to one realization of the smooth simply-connected 4-manifolds $X(m, n)$ described in Subsection 4.1

realize this desingularized double branched cover as the quotient by $\iota \times \iota$ of the blow-up of $\Sigma_{m-1} \times \Sigma_{n-1}$ at the fixed points of $\iota \times \iota$ using the map from this quotient to $\mathbb{P}^1 \times \mathbb{P}^1$ induced by the map

$$\Sigma_{m-1} \times \Sigma_{n-1} \rightarrow (\Sigma_{m-1}/\iota) \times (\Sigma_{n-1}/\iota) \cong \mathbb{P}^1 \times \mathbb{P}^1.$$

We describe a pair of involutions of $X(m, n)$ of the form described in Proposition 2.1 from this perspective after developing a general framework for the production of such pairs on smooth complete intersections in products of complex projective spaces in Subsection 4.3.

Taking p_1 and p_2 as in (7), the polynomials cutting out the complex hypersurfaces $X_{2,m,n}$ exhibit symmetries which we show can be much more broadly realized in Subsection 4.2.

4.2. Symmetric multihomogeneous cutting polynomials. In this subsection, we prove Lemma 4.2, which shows that any smooth complete intersection cut out in a product of complex projective spaces by polynomials whose multidegrees satisfy a certain constraint is diffeomorphic to a complete intersection cut out in that product of complex projective spaces by polynomials exhibiting certain symmetries used in Subsection 4.3 to define the involutions whose existence Proposition 4.1 communicates.

Lemma 4.2. *Consider a tuple $\mathbf{n} := (n_0, \dots, n_\nu)$ of positive integers, a positive integer $m < n := n_0 + \dots + n_\nu$, and a $m \times (\nu + 1)$ matrix d of positive integers satisfying*

$$(8) \quad m_0 = 0 \quad \text{or} \quad m - 2m_0 \geq n - 2n_0$$

for $m_0 < n_0$ the number of odd entries of the 0th column of d . For a smooth complete intersection X cut out in the product $\mathbb{P}^n$ of complex projective spaces by polynomials of multidegrees d , a generic choice of multihomogeneous polynomials of multidegrees d with only real coefficients whose constituent monomials each have even power of $z_{0,0}$ and positive power of some coordinate function other than any of $z_{0,n_0}, \dots, z_{\nu,n_\nu}$ will cut out a complete intersection in $\mathbb{P}^n$ diffeomorphic to X .

Proof. With notation as in the lemma statement, we write $d =: (d_1, \dots, d_m)$ for $d_i := (d_{i,0}, \dots, d_{i,\nu})$, so d_i is the multidegree of the i th polynomial cutting

out X . We then set $Q := Q_1 \times \cdots \times Q_m$ for Q_i the complex vector space of multihomogeneous polynomials of multidegree d_i on $\mathbb{C}^{n_0+1} \times \cdots \times \mathbb{C}^{n_\nu+1}$ whose constituent monomials each have even power of $z_{0,0}$ and positive power of some coordinate function other than any of $z_{0,n_0}, \dots, z_{\nu,n_\nu}$. For a subset S of $\mathbb{P}^n$, we denote by $D(S)$ the set of $q \in Q$ such that the subset of $\mathbb{P}^n$ cut out by q has a singular point in S . Arguments identical to those applied in the proof of Lemma 3.2 show that we may complete the proof by showing that $D := D(\mathbb{P}^n)$ is a proper subset of Q . We then set

$$\begin{aligned} A &:= \{[a_0] \in \mathbb{P}^{n_0} \mid a_{0,1} = \cdots = a_{0,n_0-1} = 0\} \times \{\alpha'\}, \quad B := \mathbb{P}^n \setminus (A \cup C), \\ C &:= \{[1 : 0 : \cdots : 0] \in \mathbb{P}^{n_0}\} \times (\mathbb{P}^{n'} \setminus \{\alpha'\}) \\ (\mathbb{P}^{n'} &:= \mathbb{P}^{n_1} \times \cdots \times \mathbb{P}^{n_\nu} \ni ([0 : \cdots : 0 : 1], \dots, [0 : \cdots : 0 : 1]) =: \alpha'), \end{aligned}$$

so we have $D = D(A) \cup D(B) \cup D(C)$. Arguments nearly identical to those presented in the proof of Lemma 3.2 can be used to show that the Zariski-closures of $D(A)$ and $D(B)$ are proper subsets of Q , so we may complete the proof by showing that $D(C)$ has empty Zariski-interior in Q (in fact, it can then be straightforwardly shown that the Zariski-closure of $D(C)$ is a proper subset of Q , but we need not formalize this).

We first assume the left-hand condition of (8) holds, so $d_{i,0}$ is even for all $i \in \{1, \dots, m\}$. For $[c]^n \in C$, we may select $\beta_\alpha \in \{0, \dots, n_\alpha\}$ satisfying $c_{\alpha,\beta_\alpha} \neq 0$ for each $\alpha \in \{1, \dots, \nu\}$. Taking

$$q_{c,i} := (q_{c,i,1}, \dots, q_{c,i,m}) \quad \text{for} \quad q_{c,i,j}(z) := \delta_{i,j} z_{0,0}^{d_{j,0}} z_{1,\beta_1}^{d_{j,1}} \cdots z_{\nu,\beta_\nu}^{d_{j,\nu}},$$

we have $q_{c,i} \in Q$ and that the set $\{q_{c,1}(c), \dots, q_{c,m}(c)\}$ spans $\mathbb{C}^m$. The proof that the Zariski-closure of $D(C)$ is a proper subset of Q when the left-hand condition of (8) holds then follows identically to the proof that the Zariski-closure of $D(B)$ is a proper subset of Q in the proof of Lemma 3.2.

Now, assume the right-hand condition of (8) holds. For convenience of notation, we reorder the rows of the $m \times (\nu + 1)$ matrix $d = (d_{i,j})$ so that $d_{1,0}, \dots, d_{m_0,0}$ are each odd and $d_{m_0+1,0}, \dots, d_{m,0}$ are each even and write

$$[(z_0, \dots, z_\nu)]^n := ([z_0], \dots, [z_\nu]) \in \mathbb{P}^n$$

for $z_i \in \mathbb{C}^{n_i+1}$. Considering

$$q' \in Q' := Q_1 \times \cdots \times Q_{m_0}, \quad q'' \in Q'' := Q_{m_0+1} \times \cdots \times Q_m,$$

the subset of $\mathbb{P}^n$ cut out by (q', q'') will not be singular at $[c]^n$ exactly when $d(q', q'')_c$ has full rank, which is certainly guaranteed when the matrices

$$(9) \quad \begin{bmatrix} \frac{\partial q'}{\partial z_{0,1}} & \cdots & \frac{\partial q'}{\partial z_{0,m_0}} \end{bmatrix} (c),$$

$$(10) \quad \begin{bmatrix} \frac{\partial q''}{\partial z_{1,0}} & \cdots & \frac{\partial q''}{\partial z_{1,n_1}} & \cdots & \frac{\partial q''}{\partial z_{\nu,0}} & \cdots & \frac{\partial q''}{\partial z_{\nu,n_\nu}} \end{bmatrix} (c)$$

have ranks m_0 and $m - m_0$ respectively. By the same arguments we applied above to complete the proof of the lemma when the left-hand condition of

(8) holds, all $q'' \in Q''$ outside of a proper Zariski-closed subset D'' of Q'' will cut out a smooth complete intersection $\overline{X}_{q''}$ in the Zariski-closure

$$\overline{C} := \{[1 : 0 : \dots : 0] \in \mathbb{P}^{n_0}\} \times \mathbb{P}^{n'}$$

of C in $\mathbb{P}^n$ whose intersection $\overline{X}_{q''} \cap C$ with C we denote by $X_{q''}$, so we note

$$(11) \quad \dim X_{q''} = \dim C - m + m_0 = n - n_0 - (m - m_0)$$

when $m - m_0 \leq n - n_0$. When $m - m_0 > n - n_0$, $X_{q''}$ is empty for any such q'' , so $D(C)$ would be contained in the proper Zariski-closed subset $Q' \times D''$ of Q . Thus, we may assume $m - m_0 \leq n - n_0$. Fixing $q'' \in Q'' \setminus D''$, the matrix (10) will then have rank $m - m_0$ for all $[c]^n \in X_{q''}$. Then, considering the map Δ taking an $m_0 \times n_0$ matrix in $\mathbb{C}^{m_0 n_0}$ to the $\binom{n_0}{m_0}$ -tuple of determinants of its $m_0 \times m_0$ minors and denoting by

$$\mathcal{D}'(X_{q''}) := \{(q', [c]^n) \in Q' \times X_{q''} \mid \Delta(\rho_c(q')) = 0\}$$

the subset of $Q' \times X_{q''}$ cut out by the homogeneous polynomials

$$(q', c) \mapsto \Delta(\rho_c(q')) \quad \text{on} \quad Q' \times \mathbb{C}^{n_1+1} \times \dots \times \mathbb{C}^{n_\nu+1},$$

we will have $(q', [c]^n) \in \mathcal{D}'(X_{q''})$ exactly when $[c]^n \in C$ is a singular point of the subset of $\mathbb{P}^n$ cut out by $(q', q'') \in Q$. As $\mathcal{D}'(X_{q''})$ is the intersection of its Zariski-closure in $Q' \times \overline{C}$ with the Zariski-open subset $Q' \times C$ of $Q' \times \overline{C}$, it is Zariski-locally-closed and thus Zariski-constructible in $Q' \times \overline{C}$. Then, considering the projection $\mathcal{D}' : Q' \times \overline{C} \rightarrow Q'$ and setting $D'(X_{q''}) := \mathcal{D}'(\mathcal{D}'(X_{q''}))$, $D'(X_{q''})$ is the image of a Zariski-constructible subset under a morphism of finite type and is thus Zariski-constructible in Q' by Chevalley's theorem [13, Exercise II.3.19]. Therefore, the dimensions of $D'(X_{q''})$ and $\mathcal{D}'(X_{q''})$ are well-defined and satisfy

$$(12) \quad \dim D'(X_{q''}) \leq \dim \mathcal{D}'(X_{q''}).$$

Then, considering the projection $\mathcal{D}' : Q' \times \overline{C} \rightarrow \overline{C}$ and noting that our above presentation of $\mathcal{D}'(X_{q''})$ shows it is Zariski-closed in the variety $Q' \times X_{q''}$, denoting by $V_1, \dots, V_v$ the finitely many irreducible components of $\mathcal{D}'(X_{q''})$, we can see that the restriction $\mathcal{D}'|_{V_i} : V_i \rightarrow \overline{\mathcal{D}'(V_i)}$ is a dominant morphism of finite-type varieties over $\mathbb{C}$ (in the language of the book of Hartshorne [13, Section I.3]) for each $i \in \{1, \dots, v\}$, where $\overline{\mathcal{D}'(V_i)}$ is the Zariski-closure of the image $\mathcal{D}'(V_i)$ of V_i under $\mathcal{D}'$. Therefore, we may apply the fiber-dimension theorem [13, Exercise II.3.22(b)] to these morphisms. Combining (12) with the resulting inequalities and (11), we then have

$$(13) \quad \dim D'(X_{q''}) \leq n - n_0 - (m - m_0) + \sup_{[c]^n \in X_{q''}} \dim \mathcal{D}'|_{\mathcal{D}'(X_{q''})}^{-1}([c]^n).$$

Fixing $[c]^n \in X_{q''}$, we can observe that

$$\mathcal{D}'|_{\mathcal{D}'(X_{q''})}^{-1}([c]^n) = \rho_c^{-1}(\Delta^{-1}(0)) \times \{[c]^n\},$$

where $\rho_c : Q' \rightarrow \mathbb{C}^{m_0 n_0}$ is the linear map taking $q' \in Q'$ to the $m_0 \times n_0$ matrix (9). We can see that ρ_c is surjective: as $[c]^n \in C$, we may select

$\beta_\alpha \in \{0, \dots, n_\alpha\}$ satisfying $c_{\alpha, \beta_\alpha} \neq 0$ for each $\alpha \in \{1, \dots, \nu\}$ such that, for at least one such α , $\beta_\alpha \neq n_\alpha$. Then, setting

$$q'_{c,i,j} := (q'_{c,i,j,1}, \dots, q'_{c,i,j,m_0}), \quad q'_{c,i,j,k}(z) := \delta_{i,k} z_{0,0}^{d_{k,0}-1} z_{0,j}^{d_{k,1}} z_{1,\beta_1} \cdots z_{\nu,\beta_\nu}^{d_{k,\nu}},$$

for $i \in \{1, \dots, m_0\}$ and $j \in \{1, \dots, n_0\}$, we can see that $q'_{c,i,j} \in Q'$ and that $\frac{\partial q'_{c,i,j,k}}{\partial z_{0,l}}(c)$ is nonzero exactly when $k = i$ and $l = j$. Therefore, with $q' := q'_{c,i,j}$, the (k, l) th entry of the matrix (9) will be nonzero exactly when $(k, l) = (i, j)$, so surjectivity is assured. This shows that

$$\dim Q' - \dim \mathbb{C}^{m_0 n_0} = \dim \text{Ker} \rho_c = \dim \mathcal{P}'|_{\mathcal{P}'(X_{q''})}^{-1}([c]^n) - \dim \Delta^{-1}(0),$$

so directly performing the linear-algebraic computation that

$$\dim \mathbb{C}^{m_0 n_0} - \dim \Delta^{-1}(0) = n_0 - m_0 + 1,$$

we have

$$\dim \mathcal{P}'|_{\mathcal{P}'(X_{q''})}^{-1}([c]^n) = \dim Q' - n_0 + m_0 - 1.$$

Combining this equality with (13) and our assumption that the right-hand condition of (8) holds, we then have that

$$\dim D'(X_{q''}) \leq \dim Q' + n - 2n_0 - (m - 2m_0) - 1 < \dim Q',$$

so $D'(X_{q''})$ has empty Zariski-interior in Q' . Now, consider a nonempty Zariski-open subset U of Q . As D'' is a proper Zariski-closed subset of Q'' , $Q' \times D''$ is a proper Zariski-closed subset of $Q' \times Q''$, so there certainly exists some $(q', q'') \in U$ not in $Q' \times D''$. Fixing such q'' , the set $U \cap (Q' \times \{q''\})$ will then be a nonempty Zariski-open subset of $Q' \times \{q''\}$ and thus cannot be contained in $D'(X_{q''}) \times \{q''\}$, which in turn contains $D(C) \cap (Q' \times \{q''\})$. Therefore, we have

$$U \cap (Q' \times \{q''\}) \not\subseteq D(C) \cap (Q' \times \{q''\})$$

and thus $U \not\subseteq D(C)$, so $D(C)$ has empty Zariski-interior in Q . As noted above, this completes the proof that $D = D(A) \cup D(B) \cup D(C)$ is a proper subset of Q , which in turn completes the proof altogether. $\square$

4.3. Involutions of complete intersections in products of complex projective spaces. Lemma 4.3 shows that any smooth complete intersection cut out in a product of complex projective spaces by polynomials exhibiting the symmetries of interest in Subsection 4.2 admits involutions of the form of interest in Proposition 4.1.

Lemma 4.3. *For any tuple $\mathbf{n} := (n_0, \dots, n_\nu)$ of positive integers and any smooth complete intersection X cut out in the product $\mathbb{P}^{\mathbf{n}}$ of complex projective spaces by polynomials with only real coefficients whose constituent*

monomials each have even power of $z_{0,0}$ and positive power of some coordinate function other than any of $z_{0,n_0}, \dots, z_{\nu,n_\nu}$, the maps

$$\begin{aligned} ([z_0], \dots, [z_\nu]) &\mapsto ([-z_{0,0} : z_{0,1} : \dots : z_{0,n_0}], [z_1], \dots, [z_\nu]), \\ ([z_0], \dots, [z_\nu]) &\mapsto ([\bar{z}_0], \dots, [\bar{z}_\nu]) \end{aligned}$$

on $\mathbb{P}^n$ respectively restrict to commuting smooth involutions a and c of X which act as the antipodal map in one complex coordinate and complex conjugation in suitable complex local coordinates about their shared fixed point

$$([0 : \dots : 0 : 1], \dots, [0 : \dots : 0 : 1]) \in X.$$

The proof of Lemma 4.3 is identical to that of Lemma 3.3, so we avoid reproducing it here. With X as in Lemma 4.3 of complex dimension at least 2, we may also observe as in Subsection 3.3 that X/a will be singular when all polynomials cutting out X have odd degree in the zeroth factor, while otherwise, a generic choice of polynomials as in Lemma 4.2 will cut out a smooth complete intersection in $\mathbb{P}^n$ which can be given the structure of a holomorphic double branched cover with compatible real structure whose branch divisor contains a real point.

As was mentioned in Subsection 4.1, taking p_1 and p_2 as in (7), the polynomials cutting out the complex hypersurfaces $X_{2,m,n}$ are as in Lemma 4.3, providing explicit descriptions of how smooth involutions of the smooth manifolds $X(m,n)$ as in the lemma arise. Recalling the hyperelliptic involution ι of the surface Σ_g of genus g alongside the construction of the smooth 4-manifold $X(m,n)$ described in Subsection 4.1 as the quotient by $\iota \times \iota$ of the blow-up of $\Sigma_{m-1} \times \Sigma_{n-1}$ at the fixed points of $\iota \times \iota$, we may also construct such involutions from this perspective. To this end, we consider an orientation-reversing involution r on Σ_g which commutes with ι such that, about each fixed point of ι , there exist complex coordinates on Σ_g such that ι and r act as the antipodal map and complex conjugation respectively (as will be true taking ι and r as in Figures 1 and 2). It is then straightforward to extend the involutions $\iota \times \text{id}$ and $r \times r$ to the blow-up of $\Sigma_{m-1} \times \Sigma_{n-1}$ at the fixed points of $\iota \times \iota$. Since the resulting involutions commute with $\iota \times \iota$, they descend to commuting smooth involutions a and c of $X(m,n)$. These involutions then satisfy the properties of interest in Proposition 2.1, where we take x to be any fixed point of $\iota \times r$.

In addition to exhibiting involutions as in Lemma 4.3 on the smooth complete intersections $X_{2,m,n}$ as in Subsection 4.1, Lemma 4.3 can be combined with Lemma 4.2 to directly give Proposition 4.1.

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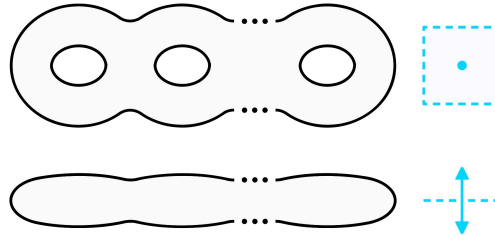


FIGURE 2. An orientation-reversing involution r of the surface Σ_g of genus g , suitably compatible with the hyperelliptic involution ι visualized in Figure 1 to allow for the construction of involutions a and c as in Lemma 4.3 on $X = X(m, n)$ from the perspective described in Subsection 4.3

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REFERENCES

- [1] M. F. Atiyah. On analytic surfaces with double points. *Proc. Roy. Soc. London Ser. A*, 247:237–244, 1958.
- [2] M. F. Atiyah and I. G. Macdonald. *Introduction to commutative algebra*. Addison-Wesley Publishing Co., Reading, Mass.-London-Don Mills, Ont., 1969.
- [3] David Baraglia and Hokuto Konno. On the Bauer-Furuta and Seiberg-Witten invariants of families of 4-manifolds. *J. Topol.*, 15(2):505–586, 2022.
- [4] David Baraglia and Hokuto Konno. A note on the Nielsen realization problem for $K3$ surfaces. *Proc. Amer. Math. Soc.*, 151(9):4079–4087, 2023.
- [5] David Baraglia and Hokuto Konno. Irreducible 4-manifolds can admit exotic diffeomorphisms, 2025.
- [6] Joan S. Birman. Abelian quotients of the mapping class group of a 2-manifold. *Bull. Amer. Math. Soc.*, 76:147–150, 1970.
- [7] Charles Ehresmann. Les connexions infinitésimales dans un espace fibré différentiable. In *Colloque de topologie (espaces fibrés), Bruxelles, 1950*, pages 29–55. Georges Thone, Liège, 1951.
- [8] Søren Galatius and Oscar Randal-Williams. Stable moduli spaces of high-dimensional manifolds. *Acta Math.*, 212(2):257–377, 2014.
- [9] Søren Galatius and Oscar Randal-Williams. Abelian quotients of mapping class groups of highly connected manifolds. *Math. Ann.*, 365(1-2):857–879, 2016.
- [10] Søren Galatius and Oscar Randal-Williams. Homological stability for moduli spaces of high dimensional manifolds. II. *Ann. of Math. (2)*, 186(1):127–204, 2017.
- [11] Robert E. Gompf and András I. Stipsicz. *4-manifolds and Kirby calculus*, volume 20 of *Graduate Studies in Mathematics*. American Mathematical Society, Providence, RI, 1999.
- [12] John L. Harer. Stability of the homology of the mapping class groups of orientable surfaces. *Ann. of Math. (2)*, 121(2):215–249, 1985.

- [13] Robin Hartshorne. *Algebraic geometry*, volume No. 52 of *Graduate Texts in Mathematics*. Springer-Verlag, New York-Heidelberg, 1977.
- [14] Sungkyung Kang, JungHwan Park, and Masaki Taniguchi. Exotic Dehn twists and homotopy coherent group actions. *Invent. Math.*, 243(1):209–241, 2026.
- [15] Hokuto Konno. Characteristic classes via 4-dimensional gauge theory. *Geom. Topol.*, 25(2):711–773, 2021.
- [16] Hokuto Konno and Jianfeng Lin. Homological instability for moduli spaces of smooth 4-manifolds, 2023.
- [17] Hokuto Konno, Jianfeng Lin, Anubhav Mukherjee, and Juan Muñoz-Echániz. The monodromy diffeomorphism of weighted singularities and Seiberg–Witten theory, 2024.
- [18] Hokuto Konno, Jianfeng Lin, Anubhav Mukherjee, and Juan Muñoz-Echániz. On four-dimensional Dehn twists and Milnor fibrations, 2024.
- [19] Hokuto Konno, Abhishek Mallick, and Masaki Taniguchi. Exotic Dehn twists on 4-manifolds, 2024.
- [20] Mustafa Korkmaz. Low-dimensional homology groups of mapping class groups: a survey. *Turkish J. Math.*, 26(1):101–114, 2002.
- [21] Manuel Krannich. Mapping class groups of highly connected $(4k + 2)$ -manifolds. *Selecta Math. (N.S.)*, 26(5):Paper No. 81, 49, 2020.
- [22] Manuel Krannich and Alexander Kupers. On Torelli groups and Dehn twists of smooth 4-manifolds. *Bull. Lond. Math. Soc.*, 57(3):956–963, 2025.
- [23] Matthias Kreck and Yang Su. Mapping class group of manifolds which look like 3-dimensional complete intersections. *Duke Math. J.*, 174(3):501–574, 2025.
- [24] P. B. Kronheimer and T. S. Mrowka. The Dehn twist on a sum of two $K3$ surfaces. *Math. Res. Lett.*, 27(6):1767–1783, 2020.
- [25] Jianfeng Lin. Isotopy of the Dehn twist on $K3 \# K3$ after a single stabilization. *Geom. Topol.*, 27(5):1987–2012, 2023.
- [26] Yujie Lin. A note on the boundary Dehn twist of $K3$ surfaces, 2025.
- [27] Ib Madsen and Michael Weiss. The stable moduli space of Riemann surfaces: Mumford’s conjecture. *Ann. of Math. (2)*, 165(3):843–941, 2007.
- [28] Edward Y. Miller. The homology of the mapping class group. *J. Differential Geom.*, 24(1):1–14, 1986.
- [29] J. Milnor. *Morse theory*, volume No. 51 of *Annals of Mathematics Studies*. Princeton University Press, Princeton, NJ, 1963. Based on lecture notes by M. Spivak and R. Wells.
- [30] Shigeyuki Morita. Characteristic classes of surface bundles. *Invent. Math.*, 90(3):551–577, 1987.
- [31] David Mumford. Abelian quotients of the Teichmüller modular group. *J. Analyse Math.*, 18:227–244, 1967.
- [32] David Mumford. Towards an enumerative geometry of the moduli space of curves. In *Arithmetic and geometry, Vol. II*, volume 36 of *Progr. Math.*, pages 271–328. Birkhäuser Boston, Boston, MA, 1983.
- [33] Patrick Orson and Mark Powell. Mapping class groups of simply connected 4-manifolds with boundary. *J. Differential Geom.*, 131(1):199–275, 2025.
- [34] Jerome Powell. Two theorems on the mapping class group of a surface. *Proc. Amer. Math. Soc.*, 68(3):347–350, 1978.
- [35] Scotty Tilton. The boundary Dehn twist on a punctured connected sum of two $K3$ surfaces is nontrivial in the smooth mapping class group, 2025.

DEPARTMENT OF MATHEMATICS, MASSACHUSETTS INSTITUTE OF TECHNOLOGY, CAMBRIDGE, MASSACHUSETTS 02139

Email address: ayodeji@mit.edu